CAPSTONE PROJECT: FINAL REPORT

Analyst: SOC Analyst
Date: April 28, 2026
Status: Complete

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1. Executive Summary

On April 28, 2026, the SOC team executed a full-scale incident response simulation. An attacker (Kali Linux, 192.168.56.170) performed reconnaissance via Nmap and exploited the Samba *usermap_script* vulnerability (CVE-2007-2447) on the Metasploitable2 target (192.168.56.140). The attacker gained **root** access. Wazuh SIEM immediately detected the activity, triggering automatic case creation in TheHive via custom integration. The case was triaged, indicators recorded, and the incident was contained. The exercise validated the entire detect–triage–respond pipeline without manual intervention.

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2. Attack Timeline (UTC)

Time	Event
17:14	Nmap scan from Kali detected by Wazuh (multiple rule 2551 alerts)
17:16	Metasploit Samba exploit successful, root shell obtained
17:18	Wazuh alerts escalated (sudo to root, PAM session opened)
23:02	TheHive case automatically created by Wazuh integration

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3. Detection & Response

- **SIEM:** Wazuh Manager triggered rule 2551 (level 10) and other rules.
- **Automation:** The `custom-thehive` integration script automatically created case #4 in TheHive with full alert details, eliminating manual case creation.
- **Triage:** In TheHive, the case was reviewed; attacker IP, target IP, and port 445 were added as observables. Tasks for containment and RCA were added and completed.
- **Containment:** The attacker IP 192.168.56.170 was blocked (simulated). Metasploitable2 was isolated (simulated).
- **RCA:** Vulnerability in Samba usermap_script (unpatched) led to remote code execution.

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4. Metrics

- **MTTD:** 0 seconds (case was created instantly upon alert)
- **MTTR:** Minutes (case creation automatic; manual triage took ~10 mins)
- **Dwell Time:** ~4 minutes (from first scan to root access)

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5. MITRE ATT&CK Mapping

Tactic	Technique
Reconnaissance	T1046 – Network Service Scanning
Exploitation	T1210 – Exploitation of Remote Services
Privilege Escalation	T1548 – Abuse Elevation Control Mechanism
Persistence	T1078 – Valid Accounts (root session)

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6. Stakeholder Briefing (150 words)

"During a routine security exercise, the SOC detected and responded to an attempted breach of a legacy server. An attacker scanned the network and exploited a known Samba vulnerability to gain root access. Our Wazuh SIEM immediately generated alerts, and our automated integration created a case in TheHive without any analyst intervention. The incident was contained within minutes, and a thorough root cause analysis was conducted. The vulnerability has been flagged for immediate patching. The successful detection and automatic response demonstrate our readiness to handle real-world attacks efficiently. We continue to improve our automated playbooks to further reduce response times."

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7. Screenshots

File	Description
nmap_scan.png	Kali nmap results against Metasploitable2
metasploit_exploit.png	Samba exploit options and root shell
wazuh_alerts.png	Wazuh JSON alerts showing attack detection
wazuh_dashboard.png	Wazuh dashboard with rule 2551 hits
thehive_case_auto.png	TheHive case #4 created automatically